

```
In [2]: import pandas as pd
```

```
In [3]: movies=pd.read_csv(r"C:\Users\aisa\OneDrive\Desktop\movie.csv")
movies
```

Out[3]:

| movieId |        | title                              | genres                                      |
|---------|--------|------------------------------------|---------------------------------------------|
| 0       | 1      | Toy Story (1995)                   | Adventure Animation Children Comedy Fantasy |
| 1       | 2      | Jumanji (1995)                     | Adventure Children Fantasy                  |
| 2       | 3      | Grumpier Old Men (1995)            | Comedy Romance                              |
| 3       | 4      | Waiting to Exhale (1995)           | Comedy Drama Romance                        |
| 4       | 5      | Father of the Bride Part II (1995) | Comedy                                      |
| ...     | ...    | ...                                | ...                                         |
| 27273   | 131254 | Kein Bund für's Leben (2007)       | Comedy                                      |
| 27274   | 131256 | Feuer, Eis & Dosenbier (2002)      | Comedy                                      |
| 27275   | 131258 | The Pirates (2014)                 | Adventure                                   |
| 27276   | 131260 | Rentun Ruusu (2001)                | (no genres listed)                          |
| 27277   | 131262 | Innocence (2014)                   | Adventure Fantasy Horror                    |

27278 rows × 3 columns

```
In [4]: movies.shape
```

Out[4]: (27278, 3)

```
In [5]: ratings=pd.read_csv(r"C:\Users\aisa\OneDrive\Desktop\rating.csv")
ratings
```

Out[5]:

|          | userId | movieId | rating | timestamp           |
|----------|--------|---------|--------|---------------------|
| 0        | 1      | 2       | 3.5    | 2005-04-02 23:53:47 |
| 1        | 1      | 29      | 3.5    | 2005-04-02 23:31:16 |
| 2        | 1      | 32      | 3.5    | 2005-04-02 23:33:39 |
| 3        | 1      | 47      | 3.5    | 2005-04-02 23:32:07 |
| 4        | 1      | 50      | 3.5    | 2005-04-02 23:29:40 |
| ...      | ...    | ...     | ...    | ...                 |
| 20000258 | 138493 | 68954   | 4.5    | 2009-11-13 15:42:00 |
| 20000259 | 138493 | 69526   | 4.5    | 2009-12-03 18:31:48 |
| 20000260 | 138493 | 69644   | 3.0    | 2009-12-07 18:10:57 |
| 20000261 | 138493 | 70286   | 5.0    | 2009-11-13 15:42:24 |
| 20000262 | 138493 | 71619   | 2.5    | 2009-10-17 20:25:36 |

20000263 rows × 4 columns

In [6]: ratings.shape

Out[6]: (20000263, 4)

In [7]: tags=pd.read\_csv(r"C:\Users\aisa\OneDrive\Desktop\tag.csv")  
tags

Out[7]:

|        | userId | movieId | tag           | timestamp           |
|--------|--------|---------|---------------|---------------------|
| 0      | 18     | 4141    | Mark Waters   | 2009-04-24 18:19:40 |
| 1      | 65     | 208     | dark hero     | 2013-05-10 01:41:18 |
| 2      | 65     | 353     | dark hero     | 2013-05-10 01:41:19 |
| 3      | 65     | 521     | noir thriller | 2013-05-10 01:39:43 |
| 4      | 65     | 592     | dark hero     | 2013-05-10 01:41:18 |
| ...    | ...    | ...     | ...           | ...                 |
| 465559 | 138446 | 55999   | dragged       | 2013-01-23 23:29:32 |
| 465560 | 138446 | 55999   | Jason Bateman | 2013-01-23 23:29:38 |
| 465561 | 138446 | 55999   | quirky        | 2013-01-23 23:29:38 |
| 465562 | 138446 | 55999   | sad           | 2013-01-23 23:29:32 |
| 465563 | 138472 | 923     | rise to power | 2007-11-02 21:12:47 |

465564 rows × 4 columns

```
In [8]: tags.shape
```

```
Out[8]: (465564, 4)
```

```
In [9]: print(movies.columns)
print(ratings.columns)
print(tags.columns)
```

```
Index(['movieId', 'title', 'genres'], dtype='object')
Index(['userId', 'movieId', 'rating', 'timestamp'], dtype='object')
Index(['userId', 'movieId', 'tag', 'timestamp'], dtype='object')
```

```
In [10]: del ratings['timestamp']
del tags['timestamp']
```

```
In [11]: print(movies.columns)
print(ratings.columns)
print(tags.columns)
```

```
Index(['movieId', 'title', 'genres'], dtype='object')
Index(['userId', 'movieId', 'rating'], dtype='object')
Index(['userId', 'movieId', 'tag'], dtype='object')
```

```
In [12]: row_0 = tags.iloc[0]
type(row_0)
```

```
Out[12]: pandas.core.series.Series
```

```
In [13]: print(row_0)
```

```
userId      18
movieId    4141
tag        Mark Waters
Name: 0, dtype: object
```

```
In [14]: row_0.index
```

```
Out[14]: Index(['userId', 'movieId', 'tag'], dtype='object')
```

```
In [20]: row_0['userId']
```

```
Out[20]: np.int64(18)
```

```
In [21]: 'rating' in row_0
```

```
Out[21]: False
```

```
In [22]: row_0.name
```

```
Out[22]: 0
```

```
In [23]: row_0=row_0.rename('firstRow')
row_0.name
```

```
Out[23]: 'firstRow'
```

## DataFrames

```
In [24]: tags.head()
```

```
Out[24]:
```

|   | userId | movieId | tag           |
|---|--------|---------|---------------|
| 0 | 18     | 4141    | Mark Waters   |
| 1 | 65     | 208     | dark hero     |
| 2 | 65     | 353     | dark hero     |
| 3 | 65     | 521     | noir thriller |
| 4 | 65     | 592     | dark hero     |

```
In [25]: tags.index
```

```
Out[25]: RangeIndex(start=0, stop=465564, step=1)
```

```
In [26]: tags.columns
```

```
Out[26]: Index(['userId', 'movieId', 'tag'], dtype='object')
```

```
In [27]: tags.iloc[ [0,11,500] ]
```

```
Out[27]:
```

|     | userId | movieId | tag               |
|-----|--------|---------|-------------------|
| 0   | 18     | 4141    | Mark Waters       |
| 11  | 65     | 1783    | noir thriller     |
| 500 | 342    | 55908   | entirely dialogue |

## Descriptive Statics

```
In [30]: ratings['rating'].describe()
```

```
Out[30]:
```

|       |              |
|-------|--------------|
| count | 2.000026e+07 |
| mean  | 3.525529e+00 |
| std   | 1.051989e+00 |
| min   | 5.000000e-01 |
| 25%   | 3.000000e+00 |
| 50%   | 3.500000e+00 |
| 75%   | 4.000000e+00 |
| max   | 5.000000e+00 |

Name: rating, dtype: float64

```
In [31]: ratings.describe()
```

```
Out[31]:
```

|       | userId       | movieId      | rating       |
|-------|--------------|--------------|--------------|
| count | 2.000026e+07 | 2.000026e+07 | 2.000026e+07 |
| mean  | 6.904587e+04 | 9.041567e+03 | 3.525529e+00 |
| std   | 4.003863e+04 | 1.978948e+04 | 1.051989e+00 |
| min   | 1.000000e+00 | 1.000000e+00 | 5.000000e-01 |
| 25%   | 3.439500e+04 | 9.020000e+02 | 3.000000e+00 |
| 50%   | 6.914100e+04 | 2.167000e+03 | 3.500000e+00 |
| 75%   | 1.036370e+05 | 4.770000e+03 | 4.000000e+00 |
| max   | 1.384930e+05 | 1.312620e+05 | 5.000000e+00 |

```
In [32]: ratings['rating'].mean()
```

```
Out[32]: np.float64(3.5255285642993797)
```

```
In [33]: ratings.mean()
```

```
Out[33]:
```

|         |              |
|---------|--------------|
| userId  | 69045.872583 |
| movieId | 9041.567330  |
| rating  | 3.525529     |
| dtype:  | float64      |

```
In [34]: ratings['rating'].min()
```

```
Out[34]: 0.5
```

```
In [35]: ratings['rating'].max()
```

```
Out[35]: 5.0
```

```
In [36]: ratings['rating'].std()
```

```
Out[36]: 1.051988919275684
```

```
In [37]: ratings['rating'].mode()
```

```
Out[37]:
```

|   |     |
|---|-----|
| 0 | 4.0 |
|---|-----|

Name: rating, dtype: float64

```
In [39]: ratings.corr()
```

Out[39]:

|                | userId    | movieId   | rating   |
|----------------|-----------|-----------|----------|
| <b>userId</b>  | 1.000000  | -0.000850 | 0.001175 |
| <b>movieId</b> | -0.000850 | 1.000000  | 0.002606 |
| <b>rating</b>  | 0.001175  | 0.002606  | 1.000000 |

```
In [40]: filter1=ratings['rating']>10
print(filter1)
filter1.any()
```

```
0      False
1      False
2      False
3      False
4      False
...
20000258  False
20000259  False
20000260  False
20000261  False
20000262  False
Name: rating, Length: 20000263, dtype: bool
```

Out[40]: np.False\_

```
In [41]: filter2=ratings['rating']>0
filter2.all()
```

Out[41]: np.True\_

## Data Cleaning:

```
In [42]: movies.shape
```

Out[42]: (27278, 3)

```
In [43]: movies.isnull().any().any()
```

Out[43]: np.False\_

```
In [44]: ratings.shape
```

Out[44]: (20000263, 3)

```
In [45]: ratings.isnull().any().any()
```

Out[45]: np.False\_

```
In [46]: tags.shape
```

```
Out[46]: (465564, 3)
```

```
In [47]: tags.isnull().any().any()
```

```
Out[47]: np.True_
```

```
In [48]: tags=tags.dropna()
```

```
In [49]: tags.isnull().any().any()
```

```
Out[49]: np.False_
```

```
In [50]: tags.shape
```

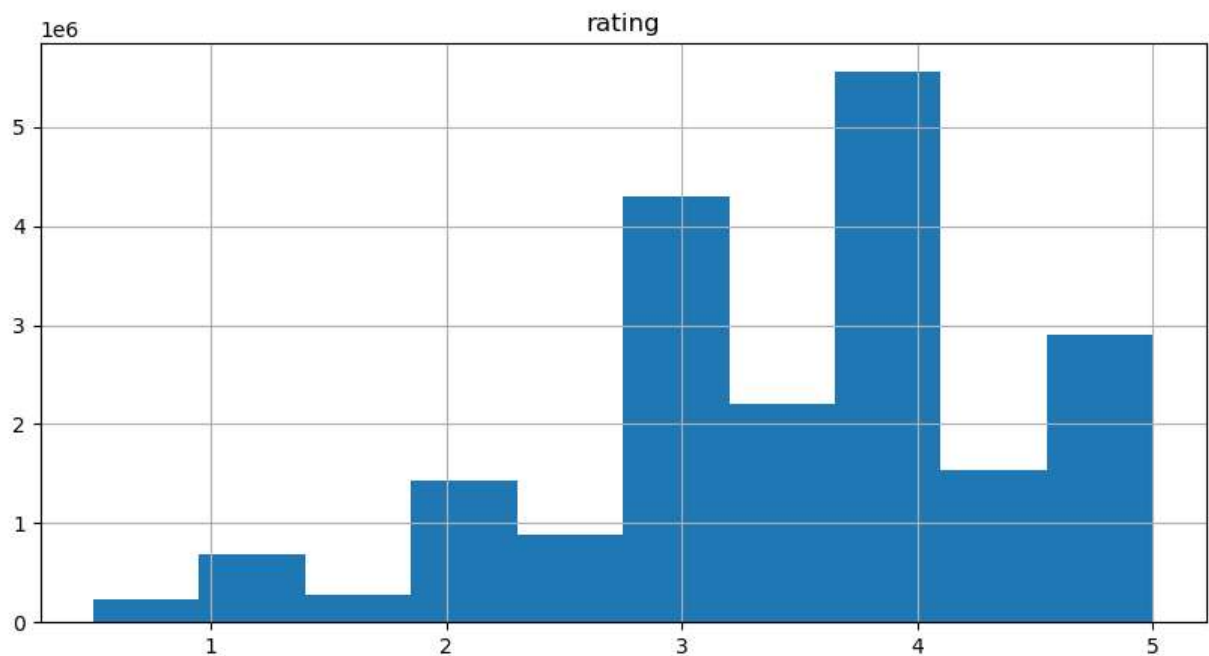
```
Out[50]: (465548, 3)
```

## Data Visualization

```
In [52]: %matplotlib inline
```

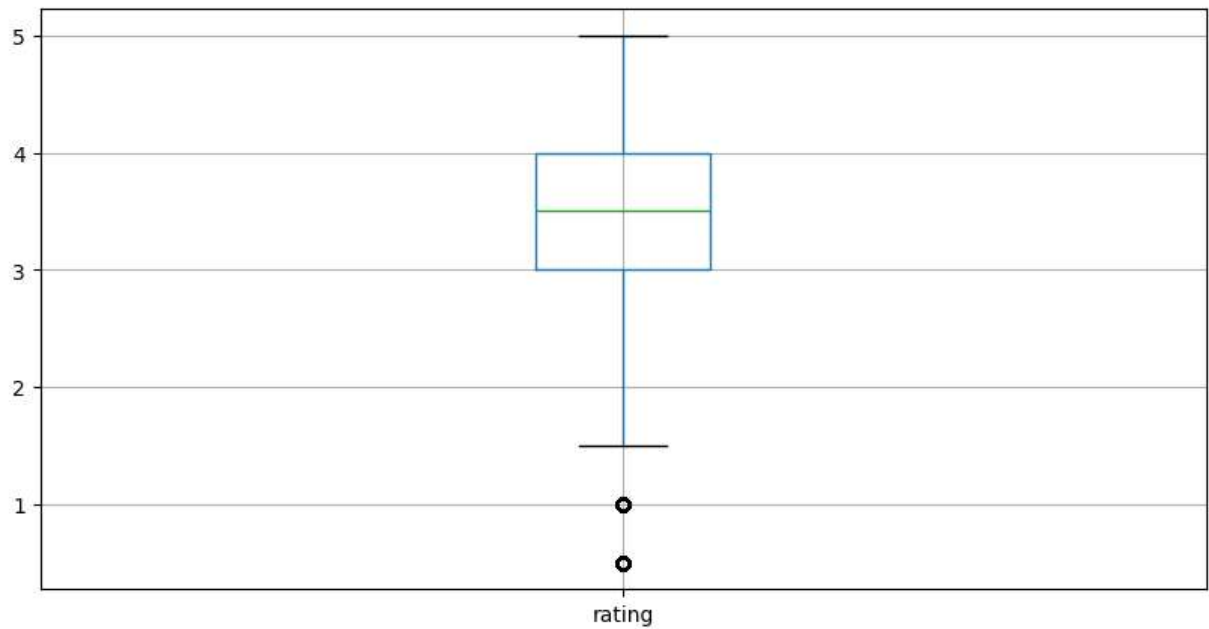
```
ratings.hist(column='rating',figsize=(10,5))
```

```
Out[52]: array([[<Axes: title={'center': 'rating'}>]], dtype=object)
```



```
In [53]: ratings.boxplot(column='rating',figsize=(10,5))
```

```
Out[53]: <Axes: >
```



# Slicing Out Columns

```
In [54]: tags['tag'].head()
```

Out[54]: 0 Mark Waters  
1 dark hero  
2 dark hero  
3 noir thriller  
4 dark hero  
Name: tag, dtype: object

```
In [55]: movies[['title', 'genres']].head()
```

Out[55]:

|   | title                              | genres                                      |
|---|------------------------------------|---------------------------------------------|
| 0 | Toy Story (1995)                   | Adventure Animation Children Comedy Fantasy |
| 1 | Jumanji (1995)                     | Adventure Children Fantasy                  |
| 2 | Grumpier Old Men (1995)            | Comedy Romance                              |
| 3 | Waiting to Exhale (1995)           | Comedy Drama Romance                        |
| 4 | Father of the Bride Part II (1995) | Comedy                                      |

```
In [56]: ratings[-10:]
```

Out[56]:

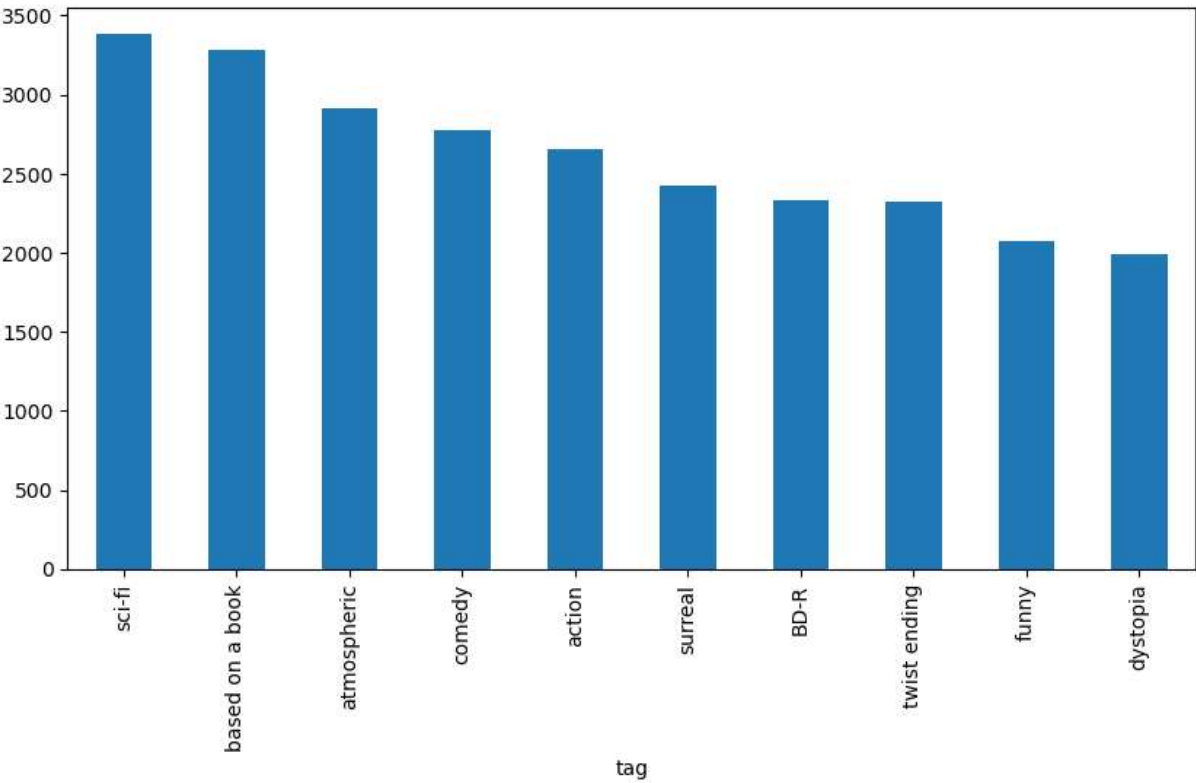
|                 | userId | movieId | rating |
|-----------------|--------|---------|--------|
| <b>20000253</b> | 138493 | 60816   | 4.5    |
| <b>20000254</b> | 138493 | 61160   | 4.0    |
| <b>20000255</b> | 138493 | 65682   | 4.5    |
| <b>20000256</b> | 138493 | 66762   | 4.5    |
| <b>20000257</b> | 138493 | 68319   | 4.5    |
| <b>20000258</b> | 138493 | 68954   | 4.5    |
| <b>20000259</b> | 138493 | 69526   | 4.5    |
| <b>20000260</b> | 138493 | 69644   | 3.0    |
| <b>20000261</b> | 138493 | 70286   | 5.0    |
| <b>20000262</b> | 138493 | 71619   | 2.5    |

```
In [57]: tag_counts=tags['tag'].value_counts()
tag_counts[-10:]
```

```
Out[57]: tag
missing child          1
Ron Moore              1
Citizen Kane          1
mullet                1
biker gang            1
Paul Adelstein        1
the wig               1
killer fish           1
genetically modified monsters  1
topless scene         1
Name: count, dtype: int64
```

```
In [58]: tag_counts[:10].plot(kind='bar',figsize=(10,5))
```

```
Out[58]: <Axes: xlabel='tag'>
```



```
In [ ]:
```